



HighTide: an Open-Source Hardware Benchmark Suite

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1. Introduction and Motivation

- Hardware and EDA benchmark suites are challenging because they are often:
 - a static snapshot of designs in fixed technologies.
 - implemented in a single design language.
 - represent a single-purpose (e.g. CTS) rather than complete RTL-to-GDS evaluation.
 - require consistent effort to update and patch.
- HighTide bridges this critical gap by establishing a fully open-source hardware benchmark suite sourced from diverse, real-world hardware designs using AI-curated design skills to create, maintain, and optimize the suite.



<https://vlsida.github.io/HighTide> <https://arxiv.org/abs/2606.04126>

2. Quick Start

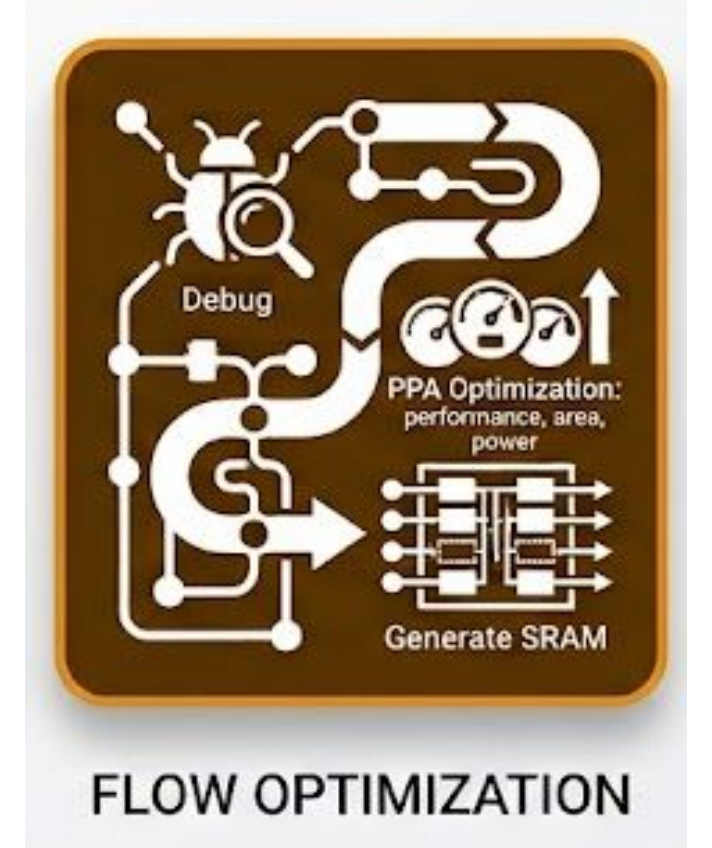
- Install Bazel
- Clone the Repository
 - <https://github.com/VLSIDA/HighTide>
- Build a Design
 - `bazel build //designs/asap7/lfsr:lfsr_final`
 - Uses cached results of others' builds
- Summarize the Results
 - `./tools/summary.sh`
- Optional: Run in OpenROAD-flow-scripts

3. AI-Assisted Design Curation

- Design Lifecycle Skills
 - Discover new designs
 - Integrate a new design
 - Port a design to a new technology
 - Track upstream design changes
- Flow Optimization Skills
 - Debug a failing design
 - Optimize Power Performance Area (PPA)
 - Generate SRAM
- Tool Reference Skills
 - OpenROAD TCL commands
 - SDC/STA commands
- Meta-Maintenance Skills
 - Track upstream bugs in tools and designs
 - Publish results for releases



DESIGN LIFECYCLE



FLOW OPTIMIZATION



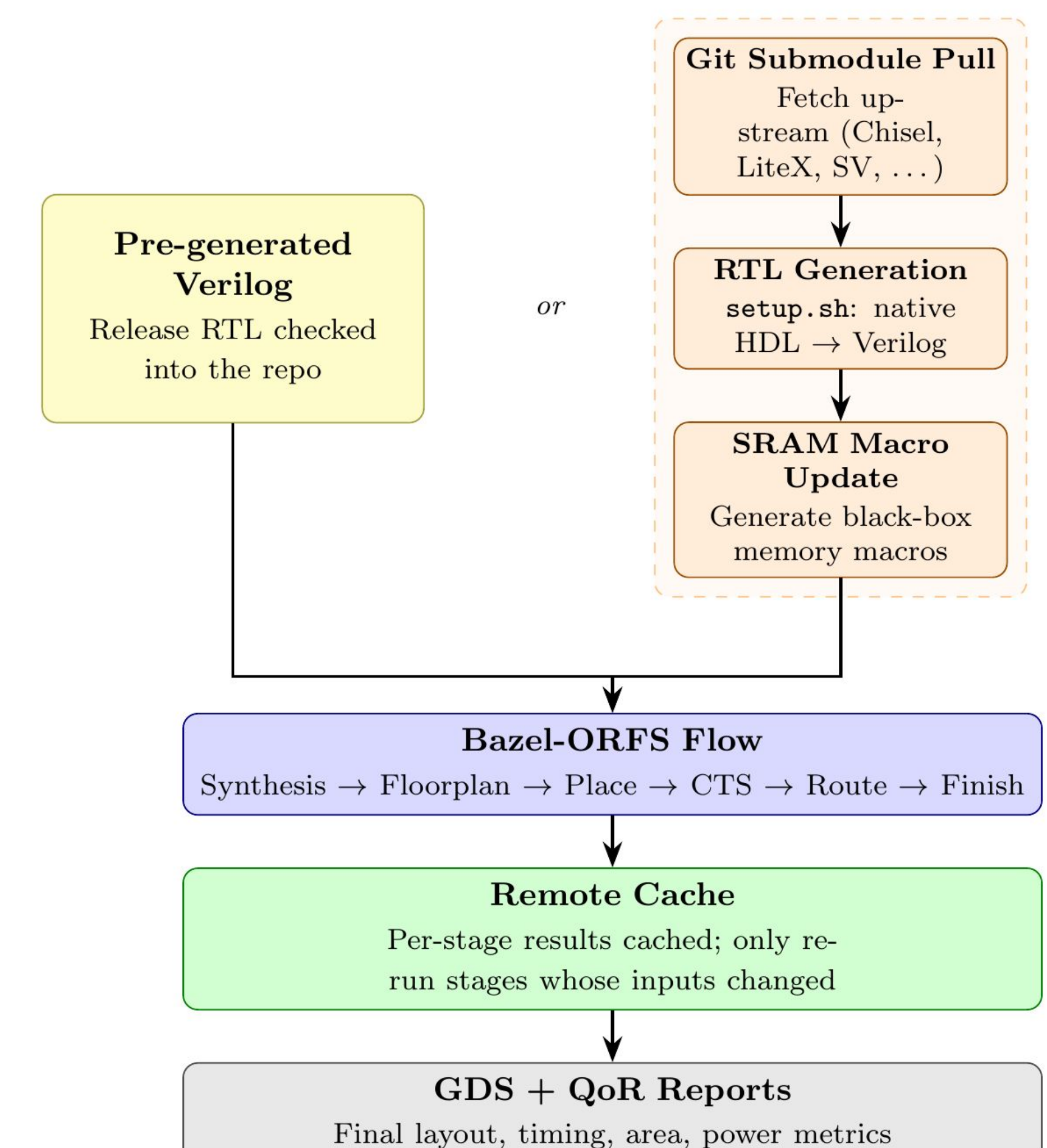
TOOL REFERENCE



META-MAINTENANCE

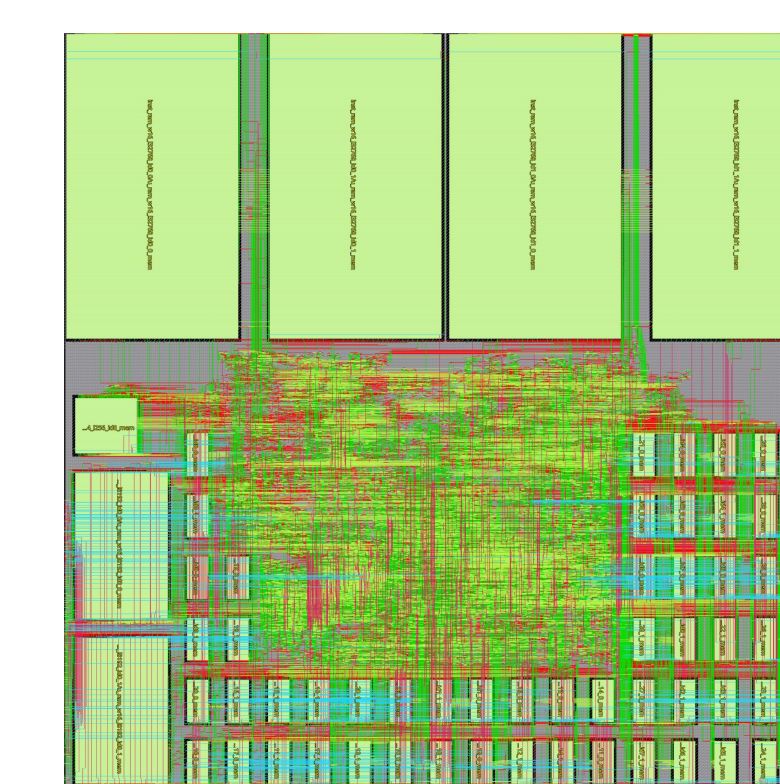
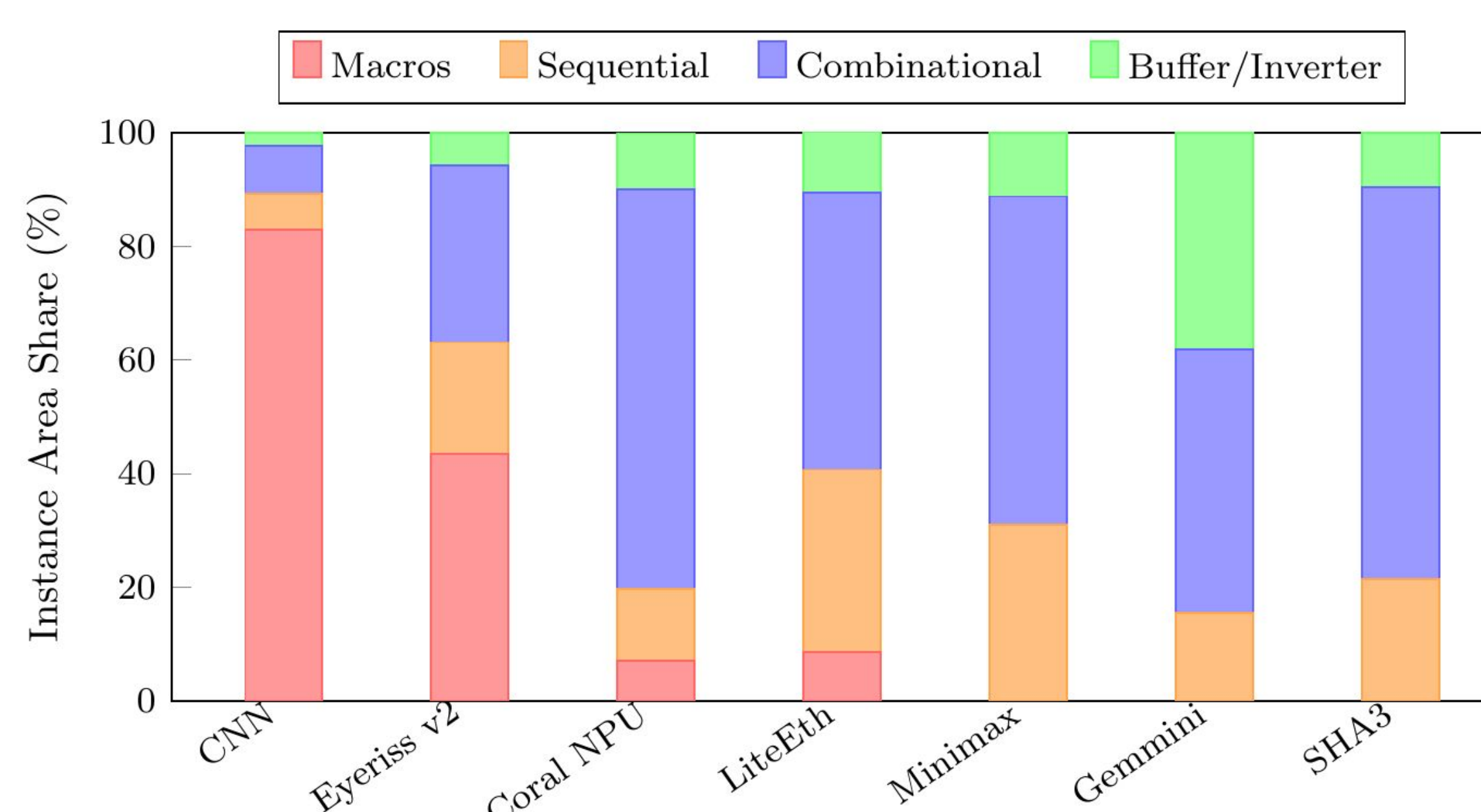
4. Benchmark Flow

- High-level languages (Chisel, Python, etc.) emit pre-generated Verilog OR saved version used
- Bazel-ORFS performs RTL-to-GDS and caches intermediate stage results both locally and remotely
- All tools and designs are pinned per HighTide release
- QoR results for releases are maintained on webpage

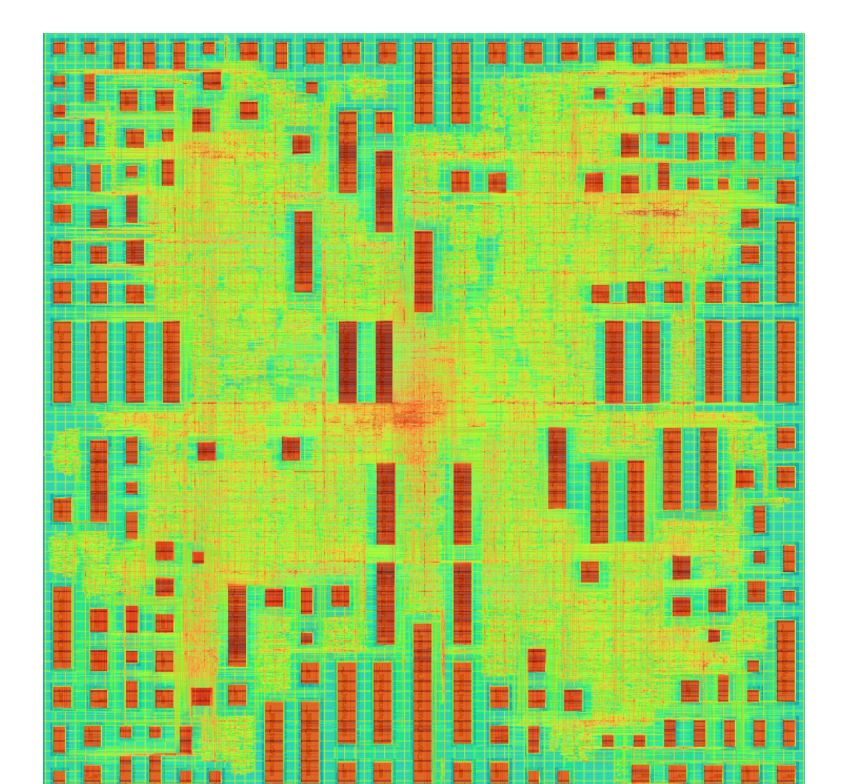


5. Results

- Over 30 Designs
 - TPUs, NPU's, GPU's, Systolic Arrays
 - 17k to 1.5M gates, up to 193 macros
- 3 Technologies (asap7, sky130, nangate45)
- Multiple Design Languages
 - Chisel, (System) Verilog, NNGen, MiGen, LiteX
- 26Q3 Tagged Release is Available
 - Plan to tag release every 3-6 months



CNN
(Python/NNGen)
ASAP7
169k cells
65 macros



Eyeriss v2
(Verilog)
ASAP7
300k cells
193 macros